

Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

Melbourne, Australia

Research Profile

Computer Science researcher and research engineer focused on **trustworthy AI security, provenance and auditability, long-horizon autonomous systems, and empirical system evaluation.**

Education

University of Melbourne

Master's degree in Computer Science

Melbourne, Australia

Expected 2026

University of British Columbia

Bachelor of Science in Computer Science

Canada

Completed Summer 2024

Selected Research and Engineering Projects

Codex Boss — *Multi-agent orchestration and evidence-based acceptance platform*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Building a platform for long-running AI workflows in which multiple agents can plan, execute, critique, and adjudicate work while preserving durable task state.
- Developed platform foundations spanning durable state/events, capability boundaries, plugin isolation, knowledge/data lifecycle, dogfooding, and acceptance/verification.
- Investigating how autonomous systems can distinguish meaningful task completion from superficially plausible but weak evidence.
- Treating unresolved failure cases as evidence for further evaluation rather than reporting an unfinished acceptance mechanism as solved.

DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building an execution layer for unattended software work with optional capabilities isolated as extensions.
- Implemented ordered/scheduled task queues, hardware-adaptive local concurrency, unified task lifecycle and terminal events, task history, and terminal notifications.
- Designed extension-failure isolation, restart/recovery-oriented execution, and an idempotent Windows installer with explicit optional-plugin flow.
- Using the system as a testbed for persistent execution, failure recovery, scheduling, and completion verification in long-running software-agent workflows.

Privacy Lens — *Trustworthy-software / privacy prototype*

- Explored privacy-risk interpretation and user-facing warning mechanisms with explicit attention to evidence boundaries, provenance, auditability, and bounded claims.
- Reinforced a research practice of separating implementation evidence from stronger security, privacy, or compliance conclusions.

Research Interests

Trustworthy AI and security; provenance and auditing; security boundaries for tool-using AI; long-horizon failure propagation; empirical evaluation of autonomous systems.

Technical and Research Strengths

- End-to-end research engineering: problem formulation, system design, iterative implementation, testing, debugging, and empirical evaluation.
- Multi-module software design with explicit persistence, lifecycle, extensibility, isolation, recovery, and acceptance criteria.
- Git/GitHub-based experimental development, branch-based iteration, regression testing, and reproducible project documentation.
- Practical multi-AI workflow design while independently reviewing plans, evidence, failure modes, and completion claims.

Current Research Direction

I am interested in **security, provenance, and auditability for tool-using AI systems**: tracing consequential outputs or actions back to the evidence and runtime state that produced them, studying failures or manipulation across long workflows, and building mechanisms that prevent model mistakes from becoming unchecked system-level actions.